Eine Woche, ein Beispiel

4.13 lattices defining abelian variety

Ref:

[Deb99]: Complex tori and abelian varieties
[Mum74]: Mumford, David, Abelian varieties. Oxford university press Oxford, 1974.
[LR22]: Herbert Lange and Rubí E. Rodríguez. Decomposition of Jacobians by Prym Varieties. 2310.
[BL04]: Christina Birkenhake, and Herbert Lange. Complex Abelian Varieties. 2nd augmented ed. Grundlehren Math. Wiss. Berlin: Springer, 2004.

This document try to work out [Deb99, p28, Ex (3)]

Claim 1. [Mum74, p35, (1) $\Leftrightarrow$ (4)]
Let $\Lambda \subseteq \mathbb{C}^g$ be a full lattice. Then

$\mathbb{C}^g/\Lambda$ is an abelian variety
$\Leftrightarrow$ $\exists$ an $\mathbb{R}$-bilinear alternating form $\omega : \mathbb{C}^g \times \mathbb{C}^g \longrightarrow \mathbb{R}$ s.t.

$$\begin{cases} \omega(\Lambda \times \Lambda) \subseteq \mathbb{Z} \\ \omega(x, ix) > 0 \quad \forall x \neq 0 \\ \omega(ix, iy) = \omega(x,y) \end{cases} \qquad (*)$$

i.e. an integral Kähler form
$$\omega(u,v) = \operatorname{Re} h(iu,v) = \operatorname{Im} h(u,v) \qquad h(au,v) = \bar{a}\, h(u,v)$$
$$= -\operatorname{Re} h(u,iv)$$

From now on. suppose $\Lambda = \langle \lambda_1, \dots, \lambda_{2g} \rangle_{\mathbb{Z}}$, we denote

$$\Pi := (a_{ij})_{i,j=1}^{2g} := (\lambda_1^*, \dots, \lambda_{2g}^*)^T \qquad \Rightarrow \lambda_i^* = \sum a_{ij} e_j^*$$

The matrix Π encodes all information of the lattice (add a basis)

Q: For what kind of conditions of Π, can we find ω satisfying $(*)$?

Let $\omega = \sum_{i<j} c_{ij}\, \lambda_i^* \wedge \lambda_j^*$, then

$$\omega(\Lambda \times \Lambda) \subset \mathbb{Z} \iff c_{ij} \in \mathbb{Z} \qquad \forall i,j.$$

Write $x = \sum x_i e_i$, $y = \sum y_i e_i$, we get

$$\begin{aligned}
\omega(x,y) &= \sum_{k,l} x_k y_l\, \omega(e_k, e_l) \\
&= \sum_{k,l} x_k y_l \sum_{i<j} c_{ij}\ \lambda_i^* \wedge \lambda_j^* (e_k, e_l) \\
&= \sum_{k,l} x_k y_l \sum_{i<j} c_{ij}\ (a_{ik} a_{jl} - a_{jk} a_{il}) \\
\omega(x, iy) &= \sum_{k,l} x_k y_{l-g} \sum_{i<j} c_{ij}\ (a_{ik} a_{jl} - a_{jk} a_{il}) \\
&= \sum_{k,l} x_k y_l \sum_{i<j} c_{ij}\ (a_{ik} a_{j(l+g)} - a_{jk} a_{i(l+g)}) \\
\omega(ix, iy) &= \sum_{k,l} x_{k-g} y_{l-g} \sum_{i<j} c_{ij}\ (a_{ik} a_{jl} - a_{jk} a_{il}) \\
&= \sum_{k,l} x_k y_l \sum_{i<j} c_{ij}\ (a_{i(k+g)} a_{j(l+g)} - a_{j(k+g)} a_{i(l+g)})
\end{aligned}$$

Therefore,

$\omega(x, ix) > 0$ is an open condition on Π.

$$\omega(ix, iy) = \omega(x,y) \iff \sum_{i<j} c_{ij} \left(\begin{vmatrix} a_{ik} & a_{il} \\ a_{jk} & a_{jl} \end{vmatrix} - \begin{vmatrix} a_{i(k+g)} & a_{i(l+g)} \\ a_{j(k+g)} & a_{j(l+g)} \end{vmatrix} \right) = 0$$

$$\forall\ k, l \in \{1, \dots, 2g\}$$

Claim 2. $\mathbb{C}^g / \Lambda$ is an abelian variety

$\Lambda = \langle \lambda_1, \dots, \lambda_{2g} \rangle_{\mathbb{Z}}$

$\Pi = (a_{ij})_{i,j=1}^{2g} = (\lambda_1^*, \dots, \lambda_{2g}^*)^T$

$\iff \exists\ c_{ij} \in \mathbb{Z}$ for all $i,j \in \{1, \dots, 2g\}$ s.t.

① $\sum_{i<j} c_{ij} \left(\begin{vmatrix} a_{ik} & a_{il} \\ a_{jk} & a_{jl} \end{vmatrix} - \begin{vmatrix} a_{i(k+g)} & a_{i(l+g)} \\ a_{j(k+g)} & a_{j(l+g)} \end{vmatrix} \right) = 0 \quad \forall k, l \in \{1, \dots, 2g\}$

② $\left(\sum_{i<j} c_{ij}\ (a_{ik} a_{j(l+g)} - a_{jk} a_{i(l+g)}) \right)_{k,l=1}^{2g}$ is def positive.

Rmk. The equations in ① are not linear independent.
There are at most $\binom{g}{2} \cdot \frac{4 \cdot 2}{4} = g(g-1)$ equations.

Reason: When $l = k$ or $l = k+g$, we get 0.
e.p. when $g = 1$, we get no condition.

When $l \neq k$, & $l \neq k+g$, denote $\{k, l, k+g, l+g\} = \{k_1, k_2, k_3, k_4\}$,

$$a_{ik_1} a_{jk_2} - a_{ik_2} a_{jk_1} - a_{ik_3} a_{jk_4} + a_{ik_4} a_{jk_3}$$

$k_1 \leftrightarrow k_2$	$k_3 \leftrightarrow k_4$	neg
$k_1 \leftrightarrow k_3$	$k_2 \leftrightarrow k_4$	neg
$k_1 \leftrightarrow k_4$	$k_2 \leftrightarrow k_3$	same

Cor. Since

$$\mathcal{A}_g \subseteq \{\mathbb{C}^n/\Lambda \text{ a.v}\} \cong \{\Pi \in \mathbb{R}^{2g \times 2g} \mid \Pi \text{ satisfies } ①② \text{ for some } (c_{ij})\} / GL_g(\mathbb{C})$$

We know

$$\dim \mathcal{A}_g = \frac{1}{2}\left((2g)^2 - g(g-1)\right) - g^2 = \frac{1}{2} g(g+1).$$

Rmk. [BL04, 4.2]

We are actually reconstructing the Riemann relations.

Let $X = \mathbb{C}^g/\Lambda$, $\quad \Lambda = \langle \lambda_1, \dots, \lambda_{2g} \rangle$, $\quad \lambda_i \in \mathbb{C}^g$

$$\Pi = (\lambda_1, \dots, \lambda_{2g})_{g\times 2g} \in M^{g\times 2g}(\mathbb{C})$$

For $\omega: \mathbb{C}^g \times \mathbb{C}^g \longrightarrow \mathbb{R}$ an nondeg alternating form, denote

$$J_D := (\omega(\lambda_i, \lambda_j))_{2g\times 2g} \in M^{2g\times 2g}(\mathbb{Z}).$$

$\omega(\Lambda \times \Lambda) \subseteq \mathbb{Z}$

Then,

① $\omega(ix, iy) = \omega(x,y) \iff \Pi J_D^{-1} \Pi^T = 0$ (RC1)

② $\omega(x, ix) > 0 \quad \forall x \neq 0 \iff i\Pi J_D^{-1} \Pi^H > 0$ (RC2)

$\Pi^H = \overline{\Pi}^T$

With the symplectic basis, $\mathcal{L} = \mathcal{L}(H, \chi)$ is of type D the Riemann relations become:

① $\Pi_2 D^{-1} \Pi_1^T - \Pi_1 D^{-1} \Pi_2^T = 0$

② $i\Pi_2 D^{-1} \Pi_1^H - i\Pi_1 D^{-1} \Pi_2^H > 0$

For $(\Pi_1, \Pi_2) = (Z, D)$, the Riemann relations become:

① $Z^T = Z$

② $\operatorname{Im} Z > 0$

Rmk. The geometrical meaning of Π:

① parametrization of cplx structures on torus;

② a "change of basis", period;

③ comparison between Betti coh & de Rham coh.

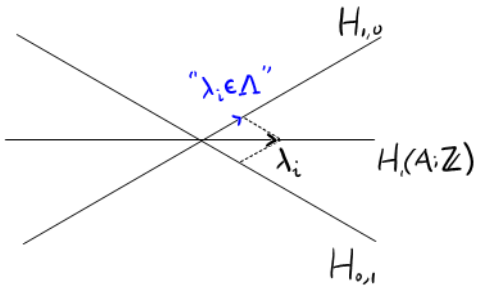
When we talk about "change of basis", it seems that they are not the basis of the same space.

$$(\lambda_1, \dots, \lambda_{2n}) = (e_1, \dots, e_n)\,\Pi$$

This is because we identify $H_1(A;\mathbb{R})$ with $H_{1,0}$ via projection. $\cong (H^{1,0})^*$

$$H_1(A;\mathbb{R}) \hookrightarrow H_1(A;\mathbb{C}) \cong H_{1,0} \oplus H_{0,1} \twoheadrightarrow H_{1,0}$$

And the actual change of basis is



$$(\lambda_1, \dots, \lambda_{2n}) = \text{“}2\,\mathrm{Re}\big((e_1, \dots, e_n)\,\Pi\big)\text{”}$$

$$= (e_1, \dots, e_n, \bar{e}_1, \dots, \bar{e}_n)\begin{pmatrix} \Pi \\ \hline \bar{\Pi} \end{pmatrix}$$

Write $\pi = \begin{pmatrix} \Pi \\ \bar{\Pi} \end{pmatrix} \overset{\text{typical}}{=\!=} \begin{pmatrix} \tau & D \\ \bar{\tau} & D \end{pmatrix}$, we have $\bar{\pi} = \begin{pmatrix} 1_n & 1_n \end{pmatrix} \pi$

$$J_\pi := \pi J_D^{-1} \pi^T = \begin{pmatrix} \Pi J_D^{-1} \Pi^T & \Pi J_D^{-1} \bar{\Pi}^H \\ \bar{\Pi} J_D^{-1} \Pi^T & \bar{\Pi} J_D^{-1} \bar{\Pi}^H \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -iM \\ iM^T & 0 \end{pmatrix} \qquad J_\pi^T = -J_\pi$$

where

$$M := i\Pi J_D^{-1} \Pi^H \qquad M > 0 \quad M^H = M$$

make it precise?

J_π is roughly the matrix of ω under the basis $\{e_1, \dots, e_n\}$. Especially,

$$(RC1) \iff H_{1,0} \text{ is isotropic subspace.}$$
$$\iff \text{Hilbert-Riemann first bilinear condition}$$
$$(RC2) \iff \text{Hilbert-Riemann second bilinear condition}$$

An explanation of Riemann Condition in terms of basis:

$$(\lambda_1, \dots, \lambda_{2n}) = (e_1, \dots, e_n, \bar{e}_1, \dots, \bar{e}_n)\pi \quad \text{in } H_1(A;\mathbb{C})$$

J acts on $H_1(A;\mathbb{R})$ by

$$J(x_1, \dots, x_n, y_1, \dots, y_n) = (x_1, \dots, x_n, y_1, \dots, y_n) J_x \qquad J_x = \begin{pmatrix} & -1_n \\ 1_n & \end{pmatrix}$$

Since $H_1(A;\mathbb{C}) = H_{1,0} \oplus H_{0,1}$, $H_{1,0}$ is the i-eigenspace of J,

$$J(e_1, \dots, e_n, \bar{e}_1, \dots, \bar{e}_n) = (e_1, \dots, e_n, \bar{e}_1, \dots, \bar{e}_n) J_e \qquad J_e = \begin{pmatrix} i1_n & \\ & -i1_n \end{pmatrix}$$

therefore

$$J(\lambda_1, \dots, \lambda_{2n}) = (\lambda_1, \dots, \lambda_{2n}) J_\lambda \qquad J_\lambda = \pi^{-1} J_e \pi$$

Automatically $J_\lambda \in M^{2n\times 2n}(\mathbb{R})$:

$$\bar{J}_\lambda = \bar{\pi}^{-1} \bar{J}_e \bar{\pi} = \pi^{-1} \begin{pmatrix} & 1_n \\ 1_n & \end{pmatrix} \begin{pmatrix} -i1_n & \\ & i1_n \end{pmatrix} \begin{pmatrix} & 1_n \\ 1_n & \end{pmatrix} \pi = J_\lambda$$

Polarization:

$$\omega \rightsquigarrow \begin{cases} J_D \in M^{2n\times 2n}(\mathbb{Z}) & \text{w.r.t. basis } \{\lambda_1, \dots, \lambda_{2n}\} \subseteq H_1(A;\mathbb{Z}) \\ J_{D,e} \in M^{2n\times 2n}(\mathbb{C}) & \text{w.r.t. basis } \{e_1, \dots, e_n, \bar{e}_1, \dots, e_n\} \subseteq H_1(A;\mathbb{C}) \\ J_{D,x} \in M^{2n\times 2n}(\mathbb{R}) & \text{w.r.t. basis } \{x_1, \dots, x_n, y_1, \dots, y_n\} \subseteq H_1(A;\mathbb{R}) \end{cases}$$

Combining with the transition matrix

$$(x_1, \dots, x_n, y_1, \dots, y_n) = (e_1, \dots, e_n, \bar{e}_1, \dots, \bar{e}_n) P_z \qquad P_z = \frac{1}{2}\begin{pmatrix} 1_n & -i1_n \\ 1_n & i1_n \end{pmatrix}$$

we know that

$$J_{D,e} = \pi^{-T} J_D \pi^{-1} \qquad J_{D,x} = P_z^T J_{D,e} P_z$$

$J_D = \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$ in actual calculation

Compatability:

$$
\begin{aligned}
& \omega(ix, iy) = \omega(x, y) \\
\Leftrightarrow\ & J_e^T J_{D,e} J_e = J_{D,e} \\
\Leftrightarrow\ & J_{D,e} = \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \\
\Leftrightarrow\ & J_{D,e}^{-1} = \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \\
\Leftrightarrow\ & \pi J_D^{-1} \pi^T = \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \\
\Leftrightarrow\ & \pi J_D^{-1} \pi^T = 0 \qquad\qquad (RC1)
\end{aligned}
$$

Now assume (RC1). Since

$$
J_{D,e}^{-1} = \pi J_D^{-1} \pi^T = \begin{pmatrix} 0 & \pi J_D^{-1} \pi^H \\ \bar{\pi} J_D^{-1} \pi^T & 0 \end{pmatrix}
$$

we write

$$
J_{D,e} = \begin{pmatrix} 0 & (\bar{\pi} J_D^{-1} \pi^T)^{-1} \\ (\pi J_D^{-1} \pi^H)^{-1} & 0 \end{pmatrix} \triangleq \begin{pmatrix} & \bar{C} \\ C & \end{pmatrix}
$$

$C^H = -C$

then

$$
\begin{aligned}
J_{D,x} &= P_z^T J_{D,e} P_z \\
&= \frac{1}{4} \begin{pmatrix} 1_n & 1_n \\ -i 1_n & i 1_n \end{pmatrix} \begin{pmatrix} 0 & \bar{C} \\ C & 0 \end{pmatrix} \begin{pmatrix} 1_n & -i 1_n \\ 1_n & i 1_n \end{pmatrix} \\
&= \frac{1}{4} \begin{pmatrix} \bar{C} + C & i\bar{C} - iC \\ -i\bar{C} + iC & \bar{C} + C \end{pmatrix} \\
&= \frac{1}{2} \begin{pmatrix} \operatorname{Re} C & \operatorname{Im} C \\ -\operatorname{Im} C & \operatorname{Re} C \end{pmatrix}
\end{aligned}
$$

Assume (RC1), one can compute Hermition with the $H_1(A;\mathbb{R})$-basis:

$$H_J(x,y) = \omega(x, iy) + i\omega(x,y) \qquad \forall x,y \in H_1(A;\mathbb{R})$$

$$\Rightarrow \quad H_J = J_{D,x} J_x + i J_{D,x}$$

$$= \frac{1}{2}\left[\begin{pmatrix} \mathrm{Re}\,C & \mathrm{Im}\,C \\ -\mathrm{Im}\,C & \mathrm{Re}\,C \end{pmatrix}\begin{pmatrix} & -1_n \\ 1_n & \end{pmatrix} + i\begin{pmatrix} \mathrm{Re}\,C & \mathrm{Im}\,C \\ -\mathrm{Im}\,C & \mathrm{Re}\,C \end{pmatrix}\right]$$

$$= \frac{1}{2}\begin{pmatrix} \mathrm{Im}\,C + i\,\mathrm{Re}\,C & -\mathrm{Re}\,C + i\,\mathrm{Im}\,C \\ \mathrm{Re}\,C - i\,\mathrm{Im}\,C & \mathrm{Im}\,C + i\,\mathrm{Re}\,C \end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix} i\overline{C} & -\overline{C} \\ \overline{C} & i\overline{C} \end{pmatrix}$$

with $\mathbb{C}_J$-basis $\{x_1, \dots, x_n\} \subseteq H_1(A;\mathbb{R})$,

$$H_{\mathbb{C}\text{-basis}} = \frac{1}{2} i \overline{C} = \frac{1}{2} i \left(\overline{\Pi} J_D^{-1} \Pi^T\right)^{-1}$$

Therefore,

$$\begin{aligned} H_J \text{ pos def} &\Leftrightarrow \tfrac{1}{2} i \left(\overline{\Pi} J_D^{-1} \Pi^T\right)^{-1} > 0 \\ &\Leftrightarrow 2i\, \overline{\Pi} J_D^{-1} \Pi^T > 0 \\ &\Leftrightarrow 2i\, \Pi J_D^{-1} \Pi^H > 0 \qquad \text{(RC2)} \end{aligned}$$